Supplementary Material

Two-Stage Bayesian Segmentation with AR(1) Errors

S1. Formal Estimation Algorithm

The following pseudocode details the sequential estimation strategy derived from the identifiability analysis.

Algorithm 1 Two-Stage Bayesian Segmentation with AR(1) Errors

Input: Time series $\mathbf{Y} = (Y_1, \ldots, Y_n)^T$, design matrices $\mathbf{X}$ (segmentation) and $\mathbf{F}$ (functional dictionary), hyper-parameters c_1, c_2, ν.

Output: Posterior estimates for structure $(\widehat{\gamma}, \widehat{r})$, coefficients $(\widehat{\beta}, \widehat{\lambda})$, variance $\widehat{\sigma}^2$, and autocorrelation $\widehat{\varphi}$.

Stage 0: Robust Initialization (Regularization)

Estimate trend: $\widehat{\mu}_0 \leftarrow \text{LOESS}(\mathbf{Y}; \text{span} = 0.1)$.
Estimate initial residuals: $\widehat{\mathbf{e}}_0 \leftarrow \mathbf{Y} - \widehat{\mu}_0$.
Initialize Autocorrelation: $\varphi^{(0)} \leftarrow \text{Corr}(\widehat{\mathbf{e}}_{0,t}, \widehat{\mathbf{e}}_{0,t-1})$. ▷ Fixes the identification metric
Initialize Indicators: $\gamma^{(0)}, r^{(0)}$ (e.g., sparse random draw or null model).

Stage 1: Structure Collapsed Gibbs Sampler

Objective: Sample from $\Pi(\gamma, r, \varphi \mid \mathbf{Y})$ *integrating out* β, λ, σ^2.
for $t = 1$ **to** $N_{burn} + N_{mcmc}$ **do**
  1. Autocorrelation Update (Metropolis-Hastings):
  Compute projected residuals $\mathbf{e}^*$ given current $\gamma^{(t-1)}, r^{(t-1)}$.
  Propose $\varphi^\star \sim \mathcal{N}(\varphi^{(t-1)}, \tau^2)$ truncated to $(-1, 1)$.
  Accept $\varphi^{(t)} \leftarrow \varphi^\star$ with probability α_{MH} based on the marginal likelihood density.
  Update precision matrices $\Sigma^{-1}_{\varphi^{(t)}}$, $\widetilde{\mathbf{T}}$, and $\widetilde{\mathbf{W}}$.
  2. Segmentation Selection (Analytic Gibbs):
  for $j = 1$ **to** d_γ **do**
    Compute Marginal Sum of Squares $S_{\gamma,r,\varphi}$ for $\gamma_j = 1$ and $\gamma_j = 0$.
    Calculate Bayes Factor BF_{10} and conditional probability $\Pi(\gamma_j = 1 \mid \gamma_{-j}, \cdot)$.
    Sample $\gamma_j^{(t)} \sim \text{Bernoulli}\left(\Pi(\gamma_j = 1 \mid \cdot)\right)$.
  end for
  3. Functional Selection (Analytic Gibbs):
  for $k = 1$ **to** d_r **do**
    Compute Marginal Sum of Squares $S_{\gamma,r,\varphi}$ for $r_k = 1$ and $r_k = 0$.
    Calculate conditional probability $\Pi(r_k = 1 \mid r_{-k}, \cdot)$.
    Sample $r_k^{(t)} \sim \text{Bernoulli}\left(\Pi(r_k = 1 \mid \cdot)\right)$.
  end for
end for

Structure Determination (Median Probability Model):
Compute marginal inclusion probabilities: $\hat{p}_j^\gamma = \frac{1}{M}\sum \gamma_j^{(t)}$ and $\hat{p}_k^r = \frac{1}{M}\sum r_k^{(t)}$.
Set active structure: $\widehat{\gamma}_j = \mathbb{I}(\hat{p}_j^\gamma > 0.5)$ and $\widehat{r}_k = \mathbb{I}(\hat{p}_k^r > 0.5)$.

Stage 2: Parameter Estimation (Sequential Identification)

Objective: Estimate coefficients conditioning on the selected model $\mathcal{M}_{\widehat{\gamma},\widehat{r}}$.
Fix design matrices $\mathbf{X}_{\widehat{\gamma}}$ and $\mathbf{F}_{\widehat{r}}$.
for $k = 1$ **to** N_{est} **do**
  Update φ**:** MH step using full conditional likelihood given $\beta^{(k-1)}, \lambda^{(k-1)}$. ▷ Conditional on stationarity
  Update σ^2**:** Sample from $\text{IG}\left(\frac{n+d_{\widehat{\gamma}}+d_{\widehat{r}}}{2}, \frac{SSR}{2}\right)$.
  Update Coefficients: Sample $(\beta^{(k)}, \lambda^{(k)})$ from Multivariate Normal block conditional on $\sigma^{2(k)}, \varphi^{(k)}$.
end for
Output: Posterior means $\widehat{\beta}, \widehat{\lambda}, \widehat{\sigma}^2, \widehat{\varphi}$ and reconstructed signal $\widehat{\mathbf{Y}} = \mathbf{X}_{\widehat{\gamma}}\widehat{\beta} + \mathbf{F}_{\widehat{r}}\widehat{\lambda}$.

S2. Algorithmic Execution Scheme

The following flowchart illustrates the coupling between the computational modules and their theoretical justification.

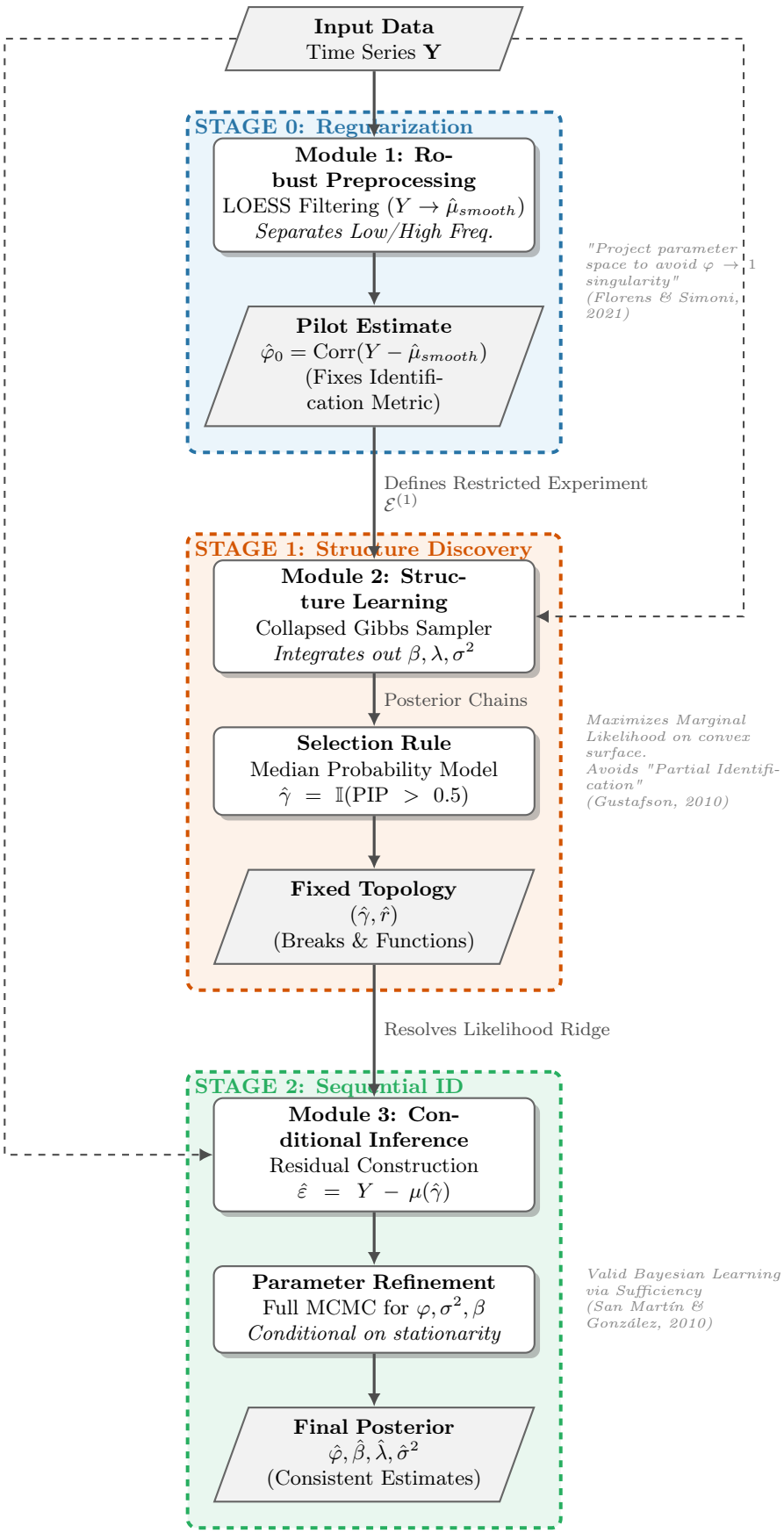

Figure 1: Algorithm Flowchart mapping computational modules to theoretical identification steps.